

Multi-Phase Flow Process (MFP) Dataset with Multi-Grained Textual Knowledge Documentation and User Guide

1. Overview

This document introduces a public multi-modal dataset for industrial process fault diagnosis, constructed from the Multi-Phase Flow Process (MFP) facility. The dataset integrates synchronized multivariate sensor measurements with structured multi-grained textual annotations, enabling joint learning from numerical signals and semantic fault knowledge.

Unlike existing industrial datasets that provide only numerical measurements, this dataset explicitly incorporates hierarchical textual descriptions of fault conditions. These descriptions span multiple semantic levels, ranging from fine-grained process-state narratives to coarse-grained fault categories and operational attributes. Such a design reflects the natural organization of industrial documentation and supports research on text-enhanced fault diagnosis, representation learning, and cross-modal reasoning.

The dataset contains 7,739 annotated samples covering one normal condition and six representative fault scenarios. Each sample consists of a windowed segment of 24-channel time-series data together with three levels of textual descriptions, forming a standardized benchmark for multi-modal industrial diagnosis.

This dataset is released to facilitate reproducible evaluation of fault diagnosis methods that leverage both sensory data and textual knowledge.

1.1. Key Features

- **Multi-modal structure:** Each sample contains aligned time-series sensor measurements and textual fault descriptions.
- **Multi-grained semantics:** Textual knowledge is organized into multi-levels, capturing both fine-grained operational conditions and coarse-grained fault categories.
- **Representative fault coverage:** The dataset includes one normal condition and six typical industrial fault scenarios.
- **Moderate scale:** A total of 7,723 samples with complete annotations are provided.
- **Reproducible benchmark:** The dataset is intended for standardized evaluation of text-enhanced fault diagnosis algorithms.

1.2. Code Repository

The accompanying code repository is available at:

<https://github.com/Kan9594/Integrating-Sensory-Data-and-Multi-Grained-Textual-Knowledge>

The dataset is released for academic research purposes only.

1.3. How to Cite

If you use this dataset in your research, please cite the following paper: *Manuscript Under Review*

2. Dataset Background

2.1. The Multi-Phase Flow Process (MFP) Facility

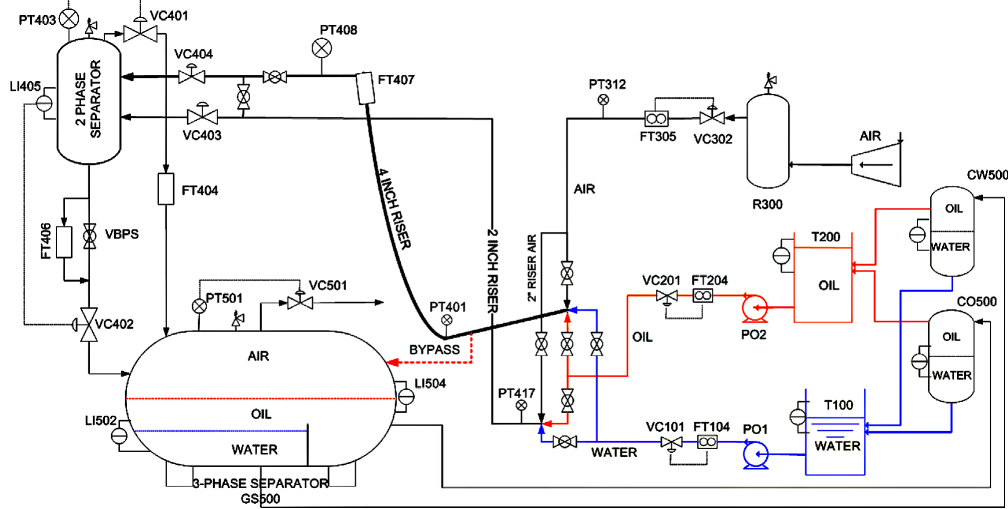


Figure 1: Diagram of the Multi-Phase Flow Process [1].

The multi-phase flow process (MFP) [1] at Cranfield University is a large-scale experimental platform for generating and controlling air–water–oil mixtures under industrially representative conditions. The MFP facility integrates independent air compressors and liquid pumps with a configurable pipeline network that includes horizontal sections, inclined flowlines, a 10.5 m riser, and both two-phase and three-phase separators. Compressed air, water, and oil are independently supplied and regulated through pumps and pneumatic valves, enabling flexible routing of the multiphase mixture through different pipeline and separation units. Twenty-four sensors provides continuous measurements of pressure, flow rate, temperature, and liquid levels at critical locations.

TABLE 1: The details of 24 measurements in the MFP.

No.	Location	Measurements	No.	Location	Measurements	No.	Location	Measurements
M1	PT312	Air delivery pressure	M9	FT104	Flow rate input water	M17	FT406	Temperature top separator output
M2	PT401	Pressure in the bottom of the riser	M10	FT407	Flow rate top riser	M18	FT104	Temperature water input
M3	PT408	Pressure in top of the riser	M11	LI405	Level top separator	M19	LI504	Level gas-liquid 3 phase separator
M4	PT403	Pressure in top separator	M12	FT406	Flow rate top separator output	M20	VC501	Position of valve VC501
M5	PT501	Pressure in 3 phase separator	M13	FT407	Density top riser	M21	VC302	Position of valve VC302
M6	PT408	Diff. pressure (PT401-PT408)	M14	FT406	Density top separator output	M22	VC101	Position of valve VC101
M7	PT403	Differential pressure over VC404	M15	FT104	Density water input	M23	PO1	Water pump current
M8	FT305	Flow rate input air	M16	FT407	Temperature top riser	M24	PT417	Pressure in mixture zone 2" line

By adjusting the water and air flow-rate setpoints, the MFP can reproduce both steady-state and dynamically varying flow regimes, enabling realistic operational variability. Moreover, through coordinated manipulation of flow-rate setpoints, valve positions, and pipeline configurations, the system can emulate a wide range of abnormal conditions. These include gradual blockages in the air or water supply lines, restrictions at the top-separator inlet, leakage-like bypassing of the riser, slugging induced by low-velocity regimes, and abnormal pressurization in isolated pipeline sections. This controllability allows the facility to be effective for the diagnostic performance evaluation.

References

- [1] C. Ruiz-Cárcel, Y. Cao, D. Mba, L. Lao, and R.T. Samuel. Statistical process monitoring of a multiphase flow facility. *Control Engineering Practice*, 42:74–88, 2015.

3. Extended Dataset Construction

3.1. Data Construction

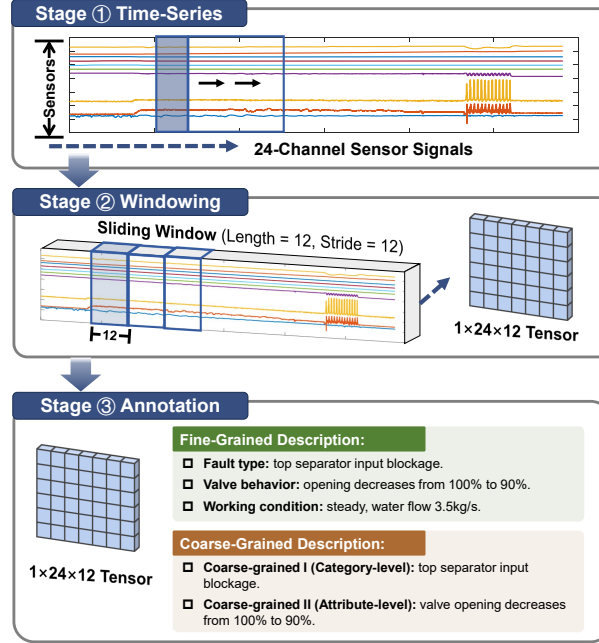


Figure 2: Construction of the multi-modal MFP dataset.

In this work, we extend existing MFP dataset by manually constructing multi-grained textual descriptions for the dataset. As illustrated in Figure 2, the extended dataset was constructed through the following process:

- 1) **Data segmentation:** A sliding window of length 12 was applied to the 24-channel time-series data, producing tensor representations $\mathbf{X}_i \in \mathbb{R}^{24 \times 24}$ for each sample.
- 2) **Text Annotation:** Domain experts created comprehensive descriptions for each sample through structured decomposition.
- 3) **Quality assurance:** All annotations were reviewed to ensure semantic consistency across samples.

Particularly, each sample is annotated with three levels of textual descriptions, forming a structured multi-grained semantic hierarchy:

- **Fine-grained description (m=0):** A complete natural-language sentence describing the fault type, valve behavior, and operating condition. This level captures detailed process semantics, including flow rates, valve positions, and system states.
- **Coarse-grained I (m=1): Fault category:** A short phrase indicating the functional-level fault type, such as “*Water line blockage*” or “*Open direct bypass*”. This level provides stable and discriminative fault identity.
- **Coarse-grained II (m=2): Operational attribute:** A concise description of the valve state or control action, such as “*valve opening decreases from 90% to 60%*”. This level encodes operational characteristics that distinguish different working conditions under the same fault type.

The multi-grained structure enables the dataset to preserve both abstract fault semantics and detailed operational context. Table 2 summarizes the designed multi-granularity textual annotations. In total, the dataset comprises 7,739 samples spanning one normal condition and six fault scenarios.

3.2. Data Structure

Each sample is stored as a JSON object with the following fields:

- **data:** a tensor of size $1 \times 24 \times 24$,
- **fine_grained:** string containing the fine-grained textual description,
- **coarse_grained_I:** string indicating the fault category,

TABLE 2: Multi-granularity textual knowledge designed for each class in the proposed dataset.

Id.	WFR (kg/s)	AFR (m ³ /s)	Description Example			Data Size
			Fine-Grained Description ($m = 0$)	Coarse ($m = 1$)	Coarse ($m = 2$)	
-	-	-	Normal, valve opening [90.1%]; normal; steady condition with water flow rate at [3.5]kg/s and air flow rate at [0.0208]m ³ /s.	Normal	Normal, valve opening [90.1%];	2313
D1-A D1-B D1-C	Vary 2 3.5	Vary 0.0417 0.0208	Air line blockage, valve opening stays at [15.1%]; Gradual closure of the air supply line manual valve to simulate progressive airflow obstruction; steady condition with water flow rate at [2.1]kg/s and air flow rate at [0.0417]m ³ /s.	Air Line blockage	Air Line blockage, valve opening stays at [15.1%]	814
D2-A D2-B D2-C	Vary 2 3.5	Vary 0.0278 0.0417	Water line blockage, valve opening decreases from [90.1%] to [60.1%]; gradual closure of the water supply line manual valve to mimic incremental liquid flow restriction; changing condition with varying water/air flow rate.	Water Line Blockage	Water line blockage, valve opening decreases from [90.1%] to [60.1%]	715
D3-A D3-B D3-C	Vary 2 3.5	Vary 0.0278 0.0208	Top separator input blockage, valve opening stays at [40.1%]; controlled restriction of the separator inlet valve VC404, replicating valve degradation effects; steady condition with water flow rate at [2.1]kg/s and air flow rate at [0.0278]m ³ /s	Top Separator Input Blockage	Top separator input blockage, valve opening stays at [40.1%]	1814
D4-A D4-B D4-C	Vary 2 3.5	Vary 0.0417 0.0208	Open direct bypass, valve opening increases from [15.1%] to [20.1%]; systematic opening of the bypass line valves to simulate leakage-induced flow diversion at the riser base; steady condition with water flow rate at [3.5]kg/s and air flow rate at [0.0208]m ³ /s.	Open Direct Bypass	Open direct bypass, valve opening increases from [15.1%] to [20.1%]	1202
D5-A D5-B	Vary Vary	Vary Vary	Slugging conditions, valve opening stays at [0.1%]; intentional reduction of air/water flow rates to induce self-sustaining pressure-flow oscillations in the riser; changing condition with varying water/air flow rate.	Slugging Conditions	Slugging conditions, valve opening stays at [0.1%]	650
D6-A D6-B	Vary Vary	Vary Vary	Pressurization of the No.2 line, valve opening stays at [0.1%]; unauthorized opening of the bridge valve connecting No.4 and No.2 lines, creating abnormal pressure accumulation in isolated piping; changing condition with varying water/air flow rate.	Pressurization of the No.2 Line	Pressurization of the No.2 line, valve opening stays at [0.1%]	231

· WFR: Water Flow Rate; AFR: Air Flow Rate.

- `coarse_grained_II`: string indicating the operational attribute.

Each line contains one complete sample and can be conveniently parsed by common programming languages such as Python.

Data Loading Example (Python)

```
import json

file_path = "dataset.jsonl"

with open(file_path, "r", encoding="utf-8") as f:
    for line in f:
        sample = json.loads(line)

        data = sample["data"] # (1, 24, 24) tensor
        fine_text = sample["fine_grained"] # fine-grained description
        coarse_I = sample["coarse_grained_I"] # fault category
        coarse_II = sample["coarse_grained_II"] # valve state
```

Each line can be read as follows:

Example of a Single Sample with Multi-Grained Annotation

Data: $[[[-1.4278, -1.364, \dots]]]$ ($1 \times 24 \times 24$ tensor)

Fine-Grained Description: {fault type: Normal; valve opening: 90.0%; fault description: Normal; working condition: steady condition with water flow rate at 3.5 kg/s and air flow rate at 0.0208 m³/s}

Coarse-Grained Description – I: {normal}

Coarse-Grained Description – II: {valve opening stays at 90.0%}